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Technical Paper

Influence of different axis-translation techniques using ceramic disks/microporous membrane filters on mechanical/hydraulic behavior of unsaturated soil

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Abstract

Element tests, such as water retention tests and triaxial tests, are efficient and widely used methods for investigating the elementary hydro-mechanical behavior of unsaturated soil. In recent research, discrepancies observed in unsaturated test results using axis-translation techniques (ATTs) with ceramic disks and microporous membrane filters (MM filters) have indicated that misunderstandings might be hiding behind the perceived elementary behavior seen in element tests. In this study, triaxial tests with ceramic disks and MM filters were firstly conducted on specimens of unsaturated completely decomposed granite sand, referred to as Masado in Japan, to compare the influence of different techniques on the test results. Then a soil–water-air coupled finite element-finite difference (FE-FD) method, based on a newly proposed unsaturated/saturated constitutive model coupled with a deformation-dependent water retention curve (WRC), was utilized to simulate triaxial tests as boundary value problems (BVPs) because of the non-uniform deformation of the specimens. Drained/vented triaxial tests showed that the stress–strain relation of Masado is basically the same regardless of the applied technique, while the amount of drainage discharge and degree of saturation during the shearing stage are quite different. The calculated average stress–strain relation basically reproduced a tendency similar to that of the tests. It was also shown from the calculation that the ATT with ceramic disks or MM filters significantly influences the distributions of saturation and stress within the specimens during shearing. It should be emphasized that unsaturated triaxial tests with the ATT are basically BVPs instead of so-called element tests.
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Keywords: Unsaturated triaxial test; Ceramic disk; Microporous membrane filter; Axis-translation technique; Soil–water-air coupling FE-FD

1. Introduction

The mechanical behavior of unsaturated soil is of major concern in geotechnical engineering and differs greatly from that of saturated soil. Different amounts of suction and degrees of saturation can significantly influence the strength and deformation of unsaturated soil, which often leads to natural disasters like the shallow-surface type of slope failure under heavy rainfall. Numerical simulations using the finite element method have been widely adopted as an effective method for understanding the mechanisms of various geotechnical problems related to unsaturated soil. Li et al. (2017) employed a modified elastoplastic constitutive model in the finite element-finite difference method and successfully described the seismic disaster process at marine sites. Xie et al. (2021) utilized a finite element program called DBLEAVES-X to reproduce the displacement

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and acceleration responses of embankment material in an unsaturated state. Moreover, the stability of slopes has also been well investigated. Slope failure in an unsaturated ground was simulated by Xiong et al. (2014) and was observed through the reproduction of its pore water/air pressure with satisfactory accuracy. A similar method was implemented by Zhang et al. (2019) to simulate a leaking underground pipe. In these research works, constitutive models for unsaturated soil were proposed and their validity was verified by element tests on both saturated and unsaturated soils, which is the key to understanding the mechanical behavior of unsaturated soil in boundary value problems. Compared with saturated soil, a quantitative evaluation of unsaturated soil is a far more complex hydraulic/mechanical coupled problem, which limits the modeling and application of element tests on unsaturated soil to engineering problems. However, its importance can never be ignored, and ceaseless effort should be given to enhance the research in this field.

Element tests, such as oedometer tests and triaxial tests, have been used to investigate the behavior of unsaturated soil under various stress paths. A critical difference in laboratory tests between unsaturated and saturated soils is the method of controlling the matric suction. The axis-translation technique (ATT), proposed by Hilf (1956), has been widely adopted as an effective method for unsaturated element tests. The main feature of the ATT is to separate air and water, commonly by a ceramic disk. Owing to the high air entry value (AEV) of the ceramic disk, many researchers have benefited from its ability to control the pore air pressure and pore water pressure independently when conducting various unsaturated laboratory tests. However, the hydraulic conductivity of the ceramic disk is approximately $10^{-8} \sim 10^{-10}$ m/s (Hong et al., 2016; Nishimura et al., 2012) and the thickness of the disk for unsaturated tests is around $4 \sim 7$ mm. The low permeability and thick depth bring about the need for an intolerably long period of time to reach the equilibrium for suction. To avoid such an unacceptably long duration, namely, up to weeks of testing, Nishimura (2012) utilized a microporous membrane filter (MM filter) in unsaturated triaxial tests to control the suction. Although the measured hydraulic conductivity of the MM filter is only slightly larger than that of the ceramic disk (Hong et al., 2016; Vaz et al., 2021; Wang et al., 2017), the thin thickness of the MM filter significantly shortens the time to reach the equilibrium for suction. The AEV of the MM filter is high enough, up to 500 kPa (Hong et al., 2016), to ensure its validity in replacing the ceramic disk in unsaturated tests. A significant reduction in the test time, achieved with the MM filter as an alternative, has been reported by many researchers, e.g., Nishimura et al. (2012). In addition, Wang et al. (2017) stated that the MM filter is advanced in controlling small suction, and in increasing and decreasing saturation. However, discrepancies between the test results using ceramic disks and MM filters in unsaturated tests have been reported for water retention tests (Hong et al., 2016) and triaxial tests (Ishikawa et al., 2010), although the mechanisms of the discrepancies of the test results have not been clarified.

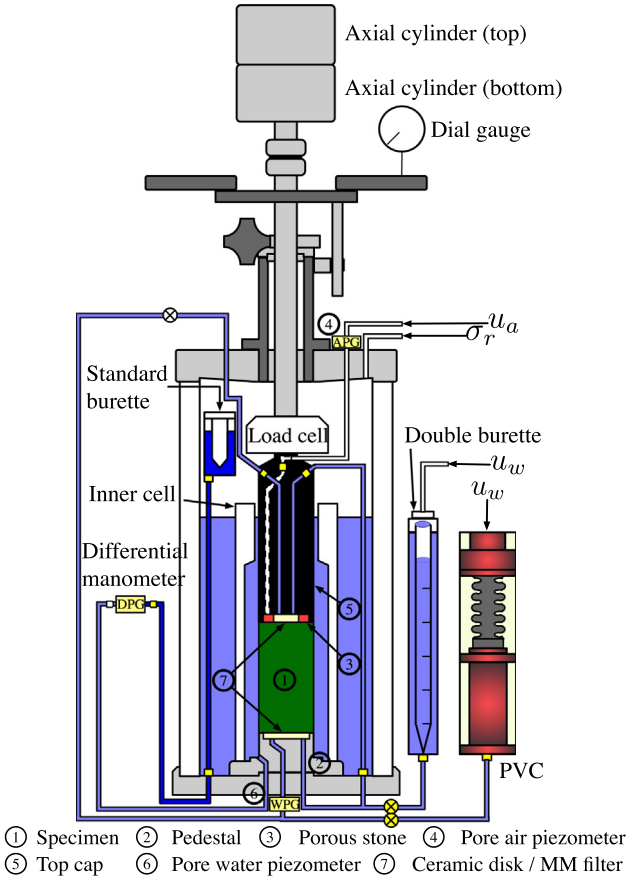


Fig. 1. Schematic diagram of triaxial apparatus.

Based on the test results using the ATT, much research has been done for the purpose of proposing a saturated/unsaturated constitutive model for soil. Zhang and Ikariya (2011) proposed a saturated/unsaturated constitutive model coupled with a water retention curve (WRC), considering the degree of saturation, S_r, as a state variable. Xiong et al. (2021) introduced the influence of deformation into the WRC model and coupled it into a saturated/unsaturated constitutive model. On the other hand, element tests are always regarded as playing an important role in investigating the soil behavior and in determining the material parameters for a constitutive model, in which the specimen is considered to behave uniformly. However, as pointed out in the work by Jin et al. (2010), element tests should not be considered capable of expressing uniform behavior in reality. The initial imperfection in a specimen and the friction between the loading cap and the specimen in element tests usually produce non-uniform soil behavior. For instance, a non-uniform strain distribution was observed in a soil specimen through X-ray CT images in a study by Watanabe et al. (2009). Noda and Yoshikawa (2015) conducted a 2D finite deformation analysis on unsaturated soil and highlighted the necessity of applying

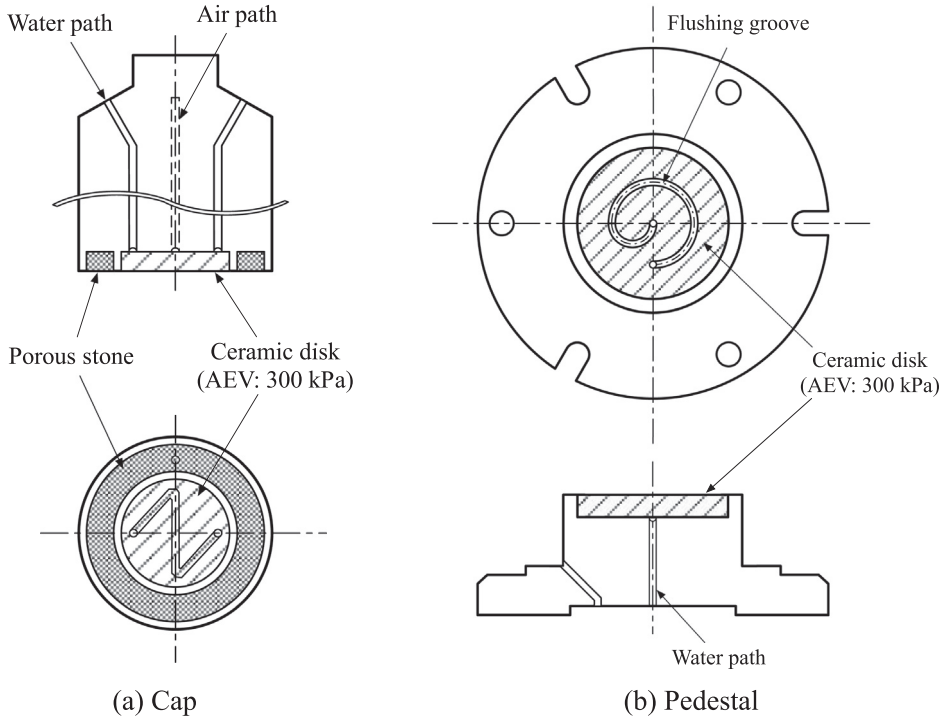


Fig. 2. Schematic diagram of cap and pedestal with ceramic disks.

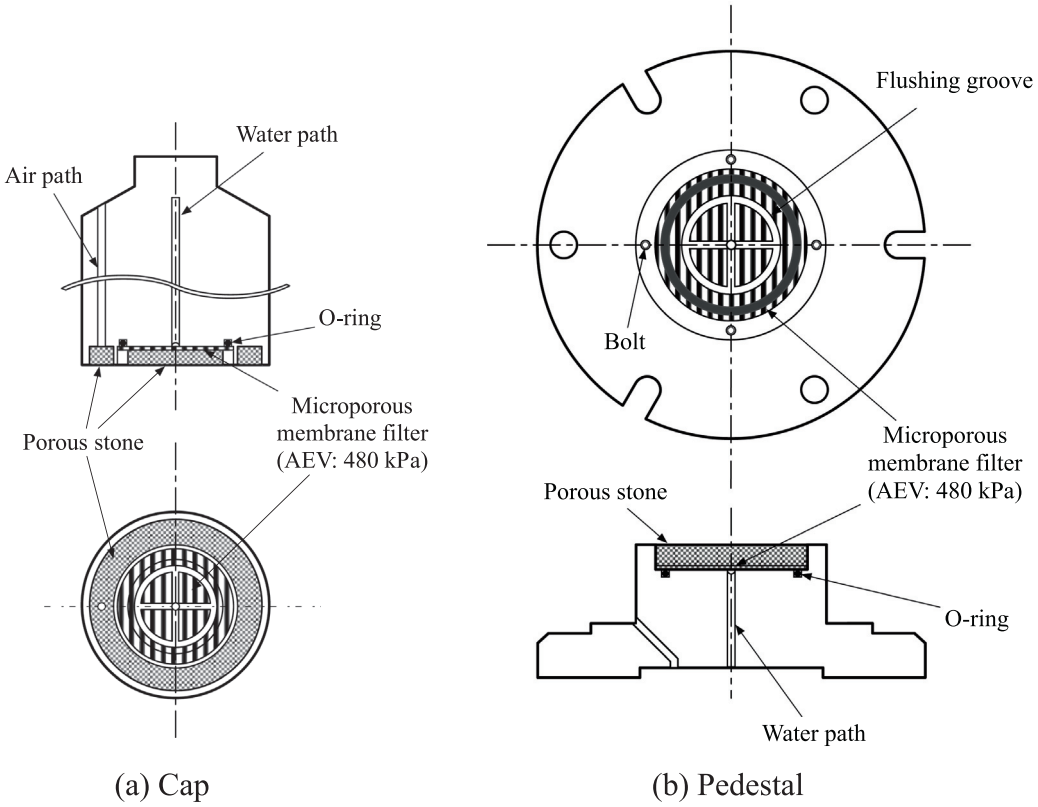


Fig. 3. Schematic diagram of cap and pedestal with MM filters.

the finite element method (FEM) to reproduce triaxial tests on unsaturated soil. Yoshikawa et al. (2015) performed a FEM simulation of triaxial tests on unsaturated silt under various drainage conditions, in which the simulated distribution of saturation varied non-uniformly within the specimen. However, these numerical simulations did not take

Table 1
Water content, void ratio, and degree of saturation at different stages of triaxial tests.

Test method	Drainage condition	Right after preparation			Before compression			End of the test		
		w_0 (%)	e_0	S_{r0}	w (%)	e	S_r	w (%)	e	S_r
Constant suction condition	Ceramic disks (Case 1)	16.5	0.631	0.696	13.2	0.576	0.613	11.8	0.575	0.574
	MM filters (Case 2)	15.7	0.605	0.691	13.6	0.554	0.652	13.9	0.584	0.634

into consideration the influence of the ceramic disks or MM filters on the test results. Later, Yoshikawa and Noda (2020) used a 2D axial symmetry mesh, with an MM filter fixed to a pedestal beneath the specimen, and calculated the consolidation process and shear process. Yoshikawa et al. (2021) substituted the MM filter in the mesh for a ceramic disk and compared the calculation results by FEM with the results of traditional triaxial tests. These calculations reproduced the test results to some degree, indicating that triaxial tests should be considered as initial/boundary problems and that the influence of ceramic disks or MM filters should not be ignored in the simulation. Nevertheless, the discrepancies between ceramic disks and MM filters have not been clearly demonstrated.

In this study, triaxial tests using the ATT with ceramic disks and MM filters were firstly conducted on specimens of a completely decomposed granite, called Masado. Then the suitable saturated/unsaturated constitutive model with a deformation-dependent WRC, proposed by Xiong et al. (2021), was utilized. Based on the constitutive model, numerical tests with soil–water-air coupling finite element numerical tests, were conducted to simulate the hydraulic and mechanical soil behavior during the triaxial tests. Finally, the results of the simulation were compared with those of the tests to understand the difference between ceramic disks and MM filters in terms of their use in the ATT.

2. Triaxial tests

2.1. Test material and apparatus

The tested soil, Masado, is widely distributed in Western Japan and is normally used as a reclaimed material for road construction. Masado is non-plastic, and its specific gravity is 2.66. The maximum dry density is 1.86 g/cm^3 and the optimum water content is 13.7 %. The detailed physical and water retention properties of Masado can be found in the work by Xiong et al. (2022b).

An unsaturated/saturated triaxial apparatus was utilized to conduct the triaxial tests on Masado. A schematic diagram of the apparatus is displayed in Fig. 1. The volumetric deformation of the specimens during each entire test was measured by a double-cell system with a differential manometer. Two sets of a top cap and a pedestal were designed for the embedment of the ceramic disks and MM filters, as seen in Figs. 2 and 3, respectively. For the set using the ceramic disks, one high AEV (300 kPa) ceramic disk was installed in the top cap and another in the

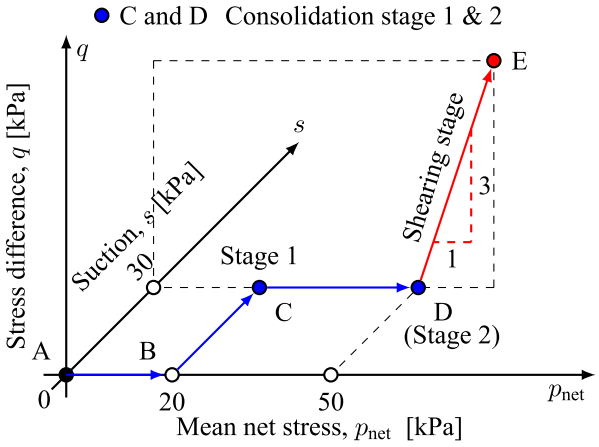
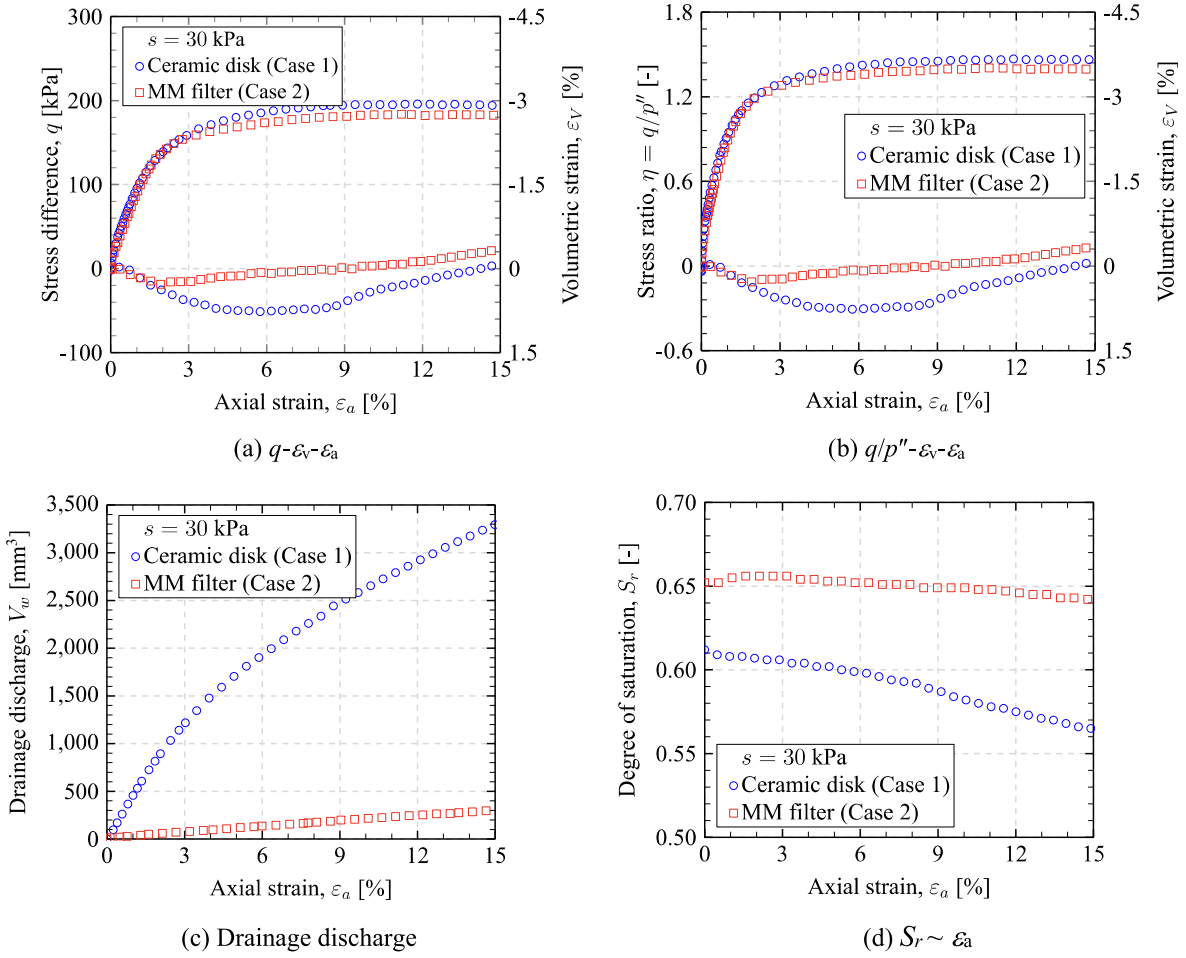


Fig. 4. Stress path in triaxial compression tests.

pedestal to fulfill dual drainage during the unsaturated tests. A porous stone was set to encircle the ceramic disk in the cap in order to apply pore air pressure. The ceramic disks were 7 mm in thickness on both sides. Meanwhile, the other set of cap and pedestal was embedded with MM filters. The MM filters used in this study were made of modified hydrophilic polyvinylidene fluoride with a thickness of 125 μm. The pore size of the filters was 0.1 μm and the AEV was 480 kPa to ensure the separation of water and air. A porous stone was also embedded in the cap to hold the filter during the controlling of the pore-air pressure. Flushing of the pore water circuit was conducted in the tests to avoid the diffusion of the air trapped in the tubes at the relatively high air pressure of >400 kPa. The triaxial tests were performed under the condition whereby no air bubbles would form in the water circuit. Moreover, only air pressure of less than 80 kPa was applied in the tests, meaning that the soil under consideration had relatively large permeability. Therefore, air diffusion was not a main issue in this research. It should be noted that one MM filter was set in the cap, while two MM filters were installed in the pedestal to enhance the tensile strength of the filters. When only one filter was set in the pedestal, small holes were observed in the filter after the triaxial tests, due to the thinness of the filter, and a strangely long duration was required to stabilize the pore water pressure. One additional filter in the pedestal was found to be sufficient to guarantee the integrity of the filters, namely, that no holes would be found after the unsaturated triaxial tests. Yet the performance of suction control during the triaxial tests



(a) q-ε_v-ε_a (b) q/p''-ε_v-ε_a

(c) Drainage discharge (d) $S_r \sim \varepsilon_a$

Fig. 5. Test results of shearing stage in unsaturated compression tests.

with two filters in the pedestal was confirmed not to have been affected by the addition of an extra filter. Thus, two filters were utilized in the pedestal during the unsaturated triaxial tests.

2.2. Sample preparation and test procedure

Two specimens of Masado for the unsaturated triaxial tests were prepared by the moisture compaction method layer by layer in three layers. The specimens were 50 mm in diameter and 100 mm in height. The initial water content of 17 % was adjusted according to the compaction test results in order to produce specimens with the relatively uniformly distributed initial void ratio of 0.65. Even though the identical procedure was adopted for the preparation of both specimens, there were still slight differences in the initial void ratio between the specimens. Table 1 presents the water content, void ratio, and degree of saturation after the specimen preparation. At the end of consolidation, the specimen with the ceramic disks had a greater void ratio, e, and smaller S_r. However, at the end of shearing, the specimen with the ceramic disks had smaller e and smaller S_r. This means that, not only the void ratio, but also the use of ceramic disks/MM filters, can influence the changes in the S_r of the soil.

Table 2
Material parameters of Masado.

Compression index $\boldsymbol{\lambda}$	0.089
Swelling index $\boldsymbol{\kappa}$	0.008
Critical state parameter $\boldsymbol{R_{cs}}$	3.65
Void ratio $\boldsymbol{N}$ (p''= 98 kPa on $N.C.L.$)	0.60
Poisson's ratio $\boldsymbol{\nu}$	0.25
Parameter of overconsolidation $\boldsymbol{a}$	10.0
Suction parameter $\boldsymbol{b}$	20.0
Parameter of overconsolidation $\boldsymbol{\beta}$	10.0
Parameter of structure $\boldsymbol{m^*}$	1.00
Void ratio $\boldsymbol{N_r}$ (p''= 98 kPa on $N.C.L.S.$)	0.63

Drained/vented triaxial tests were conducted on each Masado specimen after two consolidation stages, as shown in Fig. 4, in which the stress path in the triaxial tests is illustrated. The suction in this figure stands for the difference between the pore air pressure and the pore water pressure. After the preparation, the specimen installed at the beginning of the test was under the stress state of zero suction and zero confining stress controlled by the equipment, as

Table 3
WRC parameters for different materials.

Material	Masado	Ceramic disks	MM filters
Saturated degree of saturation S_r^s	0.75	1.0	1.0
Residual degree of saturation S_r^r	0.37	0.1	0.1
Drying AEV (kPa) S_d	6.00	300	480
Wetting AEV (kPa) S_w	1.00	100	430
Shape function parameter c_1	0.011	0.021	0.021
Shape function parameter c_2	0.042	0.042	0.032
Scanning curve parameter I_d (kPa)	1.6	1.6	1.6
Scanning curve parameter Iw (kPa)	1.1	1.1	1.1
Influence of deformation on degree of saturation on compression c_e^C	0.3	–	–
Influence of deformation on degree of saturation on dilatancy c_e^D	1.0	–	–

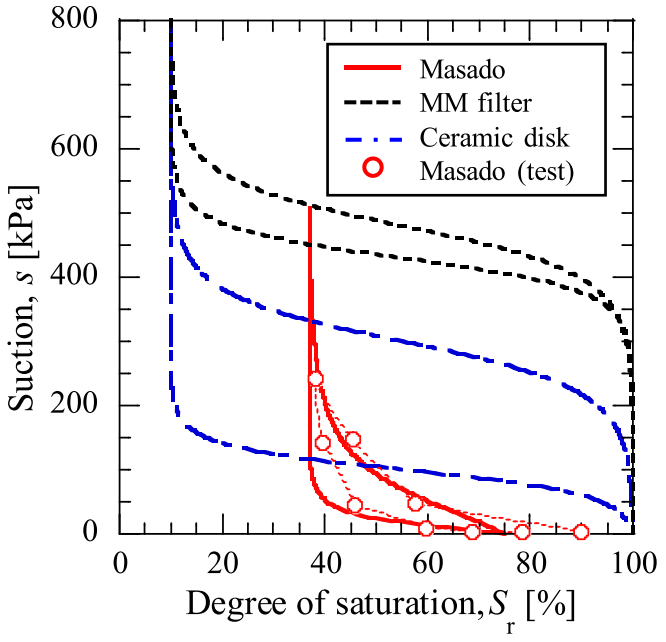


Fig. 6. Calculated water retention curves of three materials.

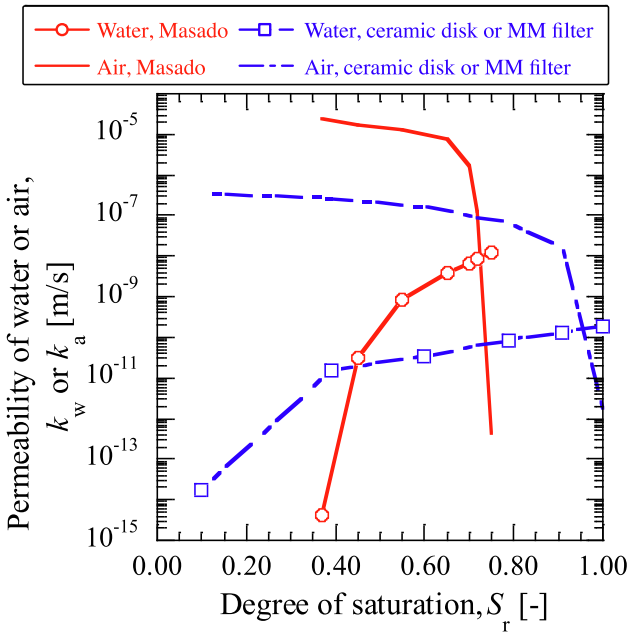


Fig. 7. Relation between permeability of water/air and degree of saturation.

represented by point A. Then the specimen was isotropically consolidated under a mean net stress, p_{net}, of 20 kPa and a suction, s, of 30 kPa (point C of Stage 1 in Fig. 4) to reach a suction equilibrium. The specimen was not loaded to Stage 2 until the change in water content was less than 0.05 %/day and the change in sample volume was ensured to be stable. A p_{net} of 50 kPa and s of 30 kPa were applied to the specimen in the second consolidation stage (point D of Stage 2 in Fig. 4). After the equalization was reached, the suction within the specimen became uniform. About 220 h and 80 h were required to reach the suction equilibrium for the ATT with the ceramic disks and MM filters, respectively. Then the specimen was sheared with a constant σ_3 under constant applied pore water and pore air pressure conditions (shearing stage in Fig. 4). The constant strain loading rate was 0.005 %/min for both specimens. Table 1 shows the water content, void ratio, and degree of saturation of the specimens before and right after compression. It should be noted that the repeatability of the test results was confirmed by conducting the test under the same conditions at least two times.

2.3. Test results

Fig. 5 illustrates the test results of the shearing stage in unsaturated compression tests. The Bishop-type effective stress was utilized to present the test results. As shown in Fig. 5(a) and (b), the stress difference ($q = \sigma_1 - \sigma_3$) and the stress ratio ($\eta = q/p''$, $p'' = (p - u_a) - S_r(u_a - u_w)$) of two triaxial tests showed convincible consistency. The stress ratio obtained in the case of the MM filters was a little smaller than that of the ceramic disks. Meanwhile, the volumetric strain, ε_V, transformed from compression to dilatancy in both tests. The phase transition took place at an axial strain (ε_a) of 2 % when using the MM filters, while it took place at an axial strain of 7 % when using the ceramic disks. Moreover, an obvious difference existed in the degree of saturation, S_r, as shown in Fig. 5(d). The variation in S_r, in the case of the ceramic disks, was much larger than that in the case of the MM filters. This was caused by the huge amount of drainage discharge of 3300 mm^3 shown

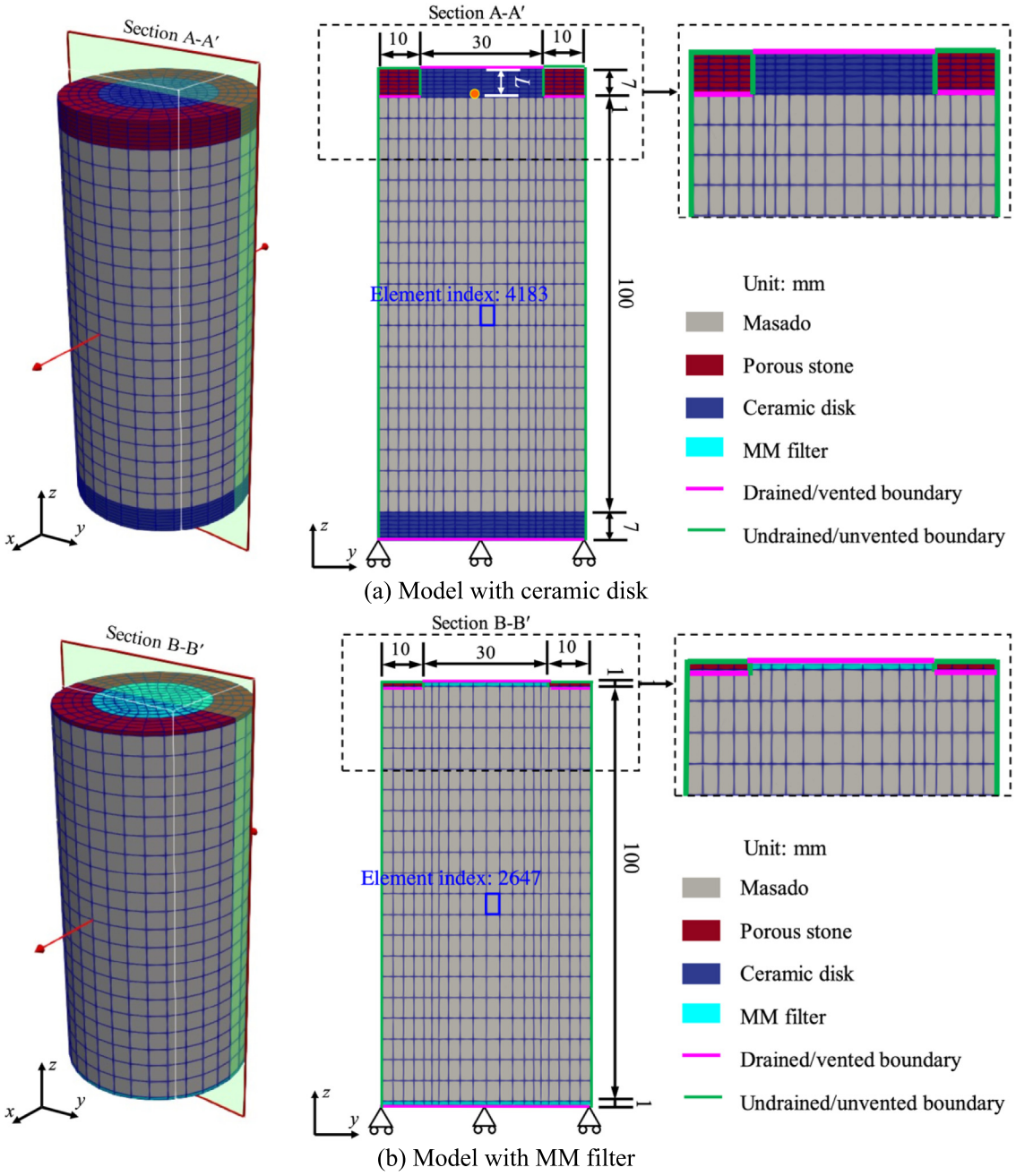
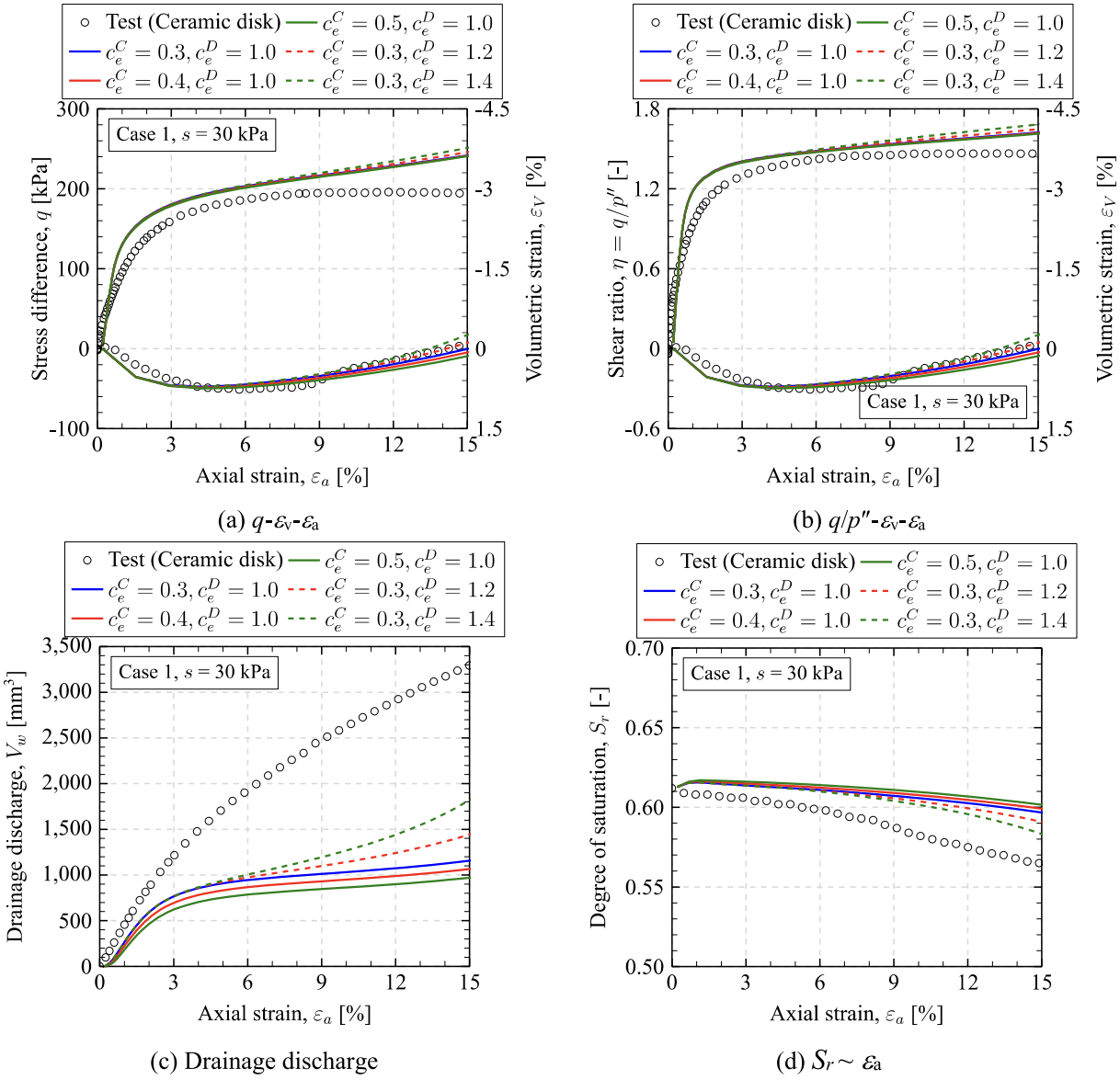


Fig. 8. Mesh and boundary conditions in FEM analyses.

in Fig. 5(c). A similar phenomenon has also been reported in other studies, e.g., Ishikawa et al. (2010). It is reasonable to assume that the different thicknesses between the ceramic disks and the MM filters caused this discrepancy. Porous ceramic is a material with water retention properties. The larger thickness and lower permeability of the ceramic disks might have generated considerable non-uniform suction inside the disks, while a similar phenomenon would not be possible with the MM filters. The suction generated in the ceramic disks increased the drainage discharge from that specimen, which also led to the long equilibrium time. These phenomena indicate that the influence of different types of ATTs should be carefully investigated and compared. Therefore, in order to understand the reason for this discrepancy, the saturated/unsaturated constitutive model proposed by Xiong et al. (2021) was utilized to conduct numerical tests in the triaxial tests on Masado, in which ceramic disks and MM filters were carefully modelled to analyze the influence of the different types of ATTs on the mechanical behavior of the soil specimens.

Table 4
Initial conditions of Masado, ceramic disks, and MM filters used in simulations.

Case	Drainage condition	Masado			Ceramic disks and MM filters		
		s_0 (kPa)	e	S_{r0}	s_0 (kPa)	e	S_{r0}
Case 1	Ceramic disks	30.0	0.576	0.613	30.0	0.515	1.00
Case 2	MM filters	30.0	0.554	0.652	30.0	–	1.00



(a) q-ε_v-ε_a (b) q/p''-ε_v-ε_a

(c) Drainage discharge (d) $S_r \sim \varepsilon_a$

Fig. 9. Influence of finite deformation parameter c_e on calculated results (Ceramic disks).

3. Numerical simulation of triaxial tests

In this chapter, to fully understand the influence of the ceramic disks and MM filters on the shearing stage, a simulation with the FE-FD method is utilized to investigate the variation inside the specimens during the tests. A saturated/unsaturated constitutive model coupled with a WRC was selected to describe the hydro-mechanical soil behavior.

3.1. Constitutive model and parameters

Based on the constitutive model proposed by Zhang and Ikariya (2011), Xiong et al. (2021) proposed a saturated/

unsaturated constitutive model coupled with a deformation-dependent WRC. Brief descriptions of the constitutive model and WRC model are provided in Appendix A and Appendix B, respectively. The traditional WRC is obtained by the relationship between suction and the degree of saturation, which does not consider the influence of the void ratio (Tarantino, 2009). Xiong et al. (2021) introduced parameter c_e to ascribe the parallel shift in the traditional intrinsic WRC to accommodate the impact of the void ratio. Satisfactory results were observed with the proposed deformation-dependent model in describing the hydro-mechanical behavior of Masado in a series of element tests (Xiong et al., 2022b). To obtain a precise description of Masado and exhibit a serious introspection into the constitutive model, the parameters for the constitutive model and WRC model in this study were identical to those in the study of Xiong et al. (2021). Tables 2 and 3 list the material parameters of Masado and the WRC parameters for the different materials, respectively.

In the numerical calculation, the ceramic disks and MM filters were set to be elastic with the same stiffness and Poisson's ratio of 20 GPa and 0.20, respectively. The porous stone had the same material parameters as the MM filters because the filter in the cap was fixed to the porous stone by screws. Moreover, the water retention properties of the ceramic disks and MM filters were considered in the numerical calculations; their parameters are provided in Table 3. Fig. 6 depicts the WRCs of Masado with the initial void ratio of 0.576, the ceramic disks, and the MM filters. Nonlinear water/air hydraulic conductivities were considered in the calculations.

Fig. 7 shows the relations between the permeability of water/air and the degree of saturation of Masado, the ceramic disks, and the MM filters used in this paper. To determine the permeability of the soil, the permeability of water at the saturated state was measured and the unsaturated permeability of water was calculated using the relative permeability coefficient of Masado (Araki et al., 2011). The relationship between the permeability of air and the degree of saturation was assumed based on the permeability of water. As the difference between the permeability of the ceramic disks and the MM filters is slight (Nishimura et al., 2012), for simplicity, the permeabilities of the two materials were set to be identical. Then, the consolidation process for the triaxial tests was calculated by the FEM and the permeability of water/air was partially adjusted by fitting the changes in the degree of saturation and the drainage discharge of the specimens, as illustrated by Xiong et al. (2022a). Finally, the values for the permeability applied in the analysis of this paper were obtained and used to produce Fig. 7.

3.2. Initial and boundary conditions

In the numerical simulations with the finite element-finite differential (FE-FD) method conducted this study, the triaxial tests were basically regarded as boundary value problems (BVPs). The numerical simulations of the triaxial tests were conducted by a program called SOFT (Xiong et al., 2014; Xiong et al., 2021). The program utilized a FE-FD scheme for dealing with soil–water-air coupling problems (Xiong et al., 2014). Fig. 8 shows the two cylindrical meshes for the cases of the ceramic disks and MM filters. L is the thickness of each ceramic disk and equals 7 mm. Even though the thickness of each MM filter was 125 μm, one element with a thickness of 1 mm was still used to model the filter and the height, while the thickness of the porous stone was modeled with several elements whose thickness was also 1 mm, and the total thickness of these elements was identical to the porous stone, as shown in Fig. 8. The porous stone between the specimen and the MM filter was ignored because of its high permeability and low water retention properties. The FEM mesh shown in Fig. 8(a) contained 9555 nodes and 8704 8-node isoparametric elements for the simulation case with the ceramic disks. For the case of the MM filters, the FEM mesh included 6279 nodes and 5632 8-node isoparametric elements.

As for the displacement boundary, the bottom surface of the FEM model was fixed only in the z-direction. The drainage/vent boundary was set identically to that in the shearing stage of the triaxial tests under constant suction conditions, while the vertical surfaces between the ceramic disk/MM filter and the porous stone were undrained/unvented.

The initial conditions of the numerical tests are presented in Table 4. The first step of the numerical tests was to set the initial void ratio, e, and S_r to be identical to the values at the beginning of the shearing stage in the triaxial tests, as shown in Table 1. Although the consolidation process could influence the distributions of suction and S_r, it was skipped, and the distributions of suction and S_r were assumed to be uniform at the beginning of the shearing stage, which allowed for an investigation into the effect of shearing on suction and S_r. The initial stress field was set to be $\sigma_{3net} = 50$ kPa. A constant axial strain load with a rate of 0.005 %/min was applied at the top surface of the model to simulate the shearing stage.

3.3. Calculation results and discussion

In this research, triaxial tests were firstly simulated with ceramic disks and MM filters, as described in Section 2. Since the deformation-dependent WRC model was adopted, the influence of parameter c_e was investigated in advance for a comparison and to gain an in-depth comprehension of the constitutive model. The calculated results are discussed from both macro and local perspectives. From the macro perspective, the calculated results are displayed using the apparent elementary behavior in the same manner as in the experiment. From the local perspective, the distributions of some physical quantities and the results of some specific elements in the numerical models are dis-

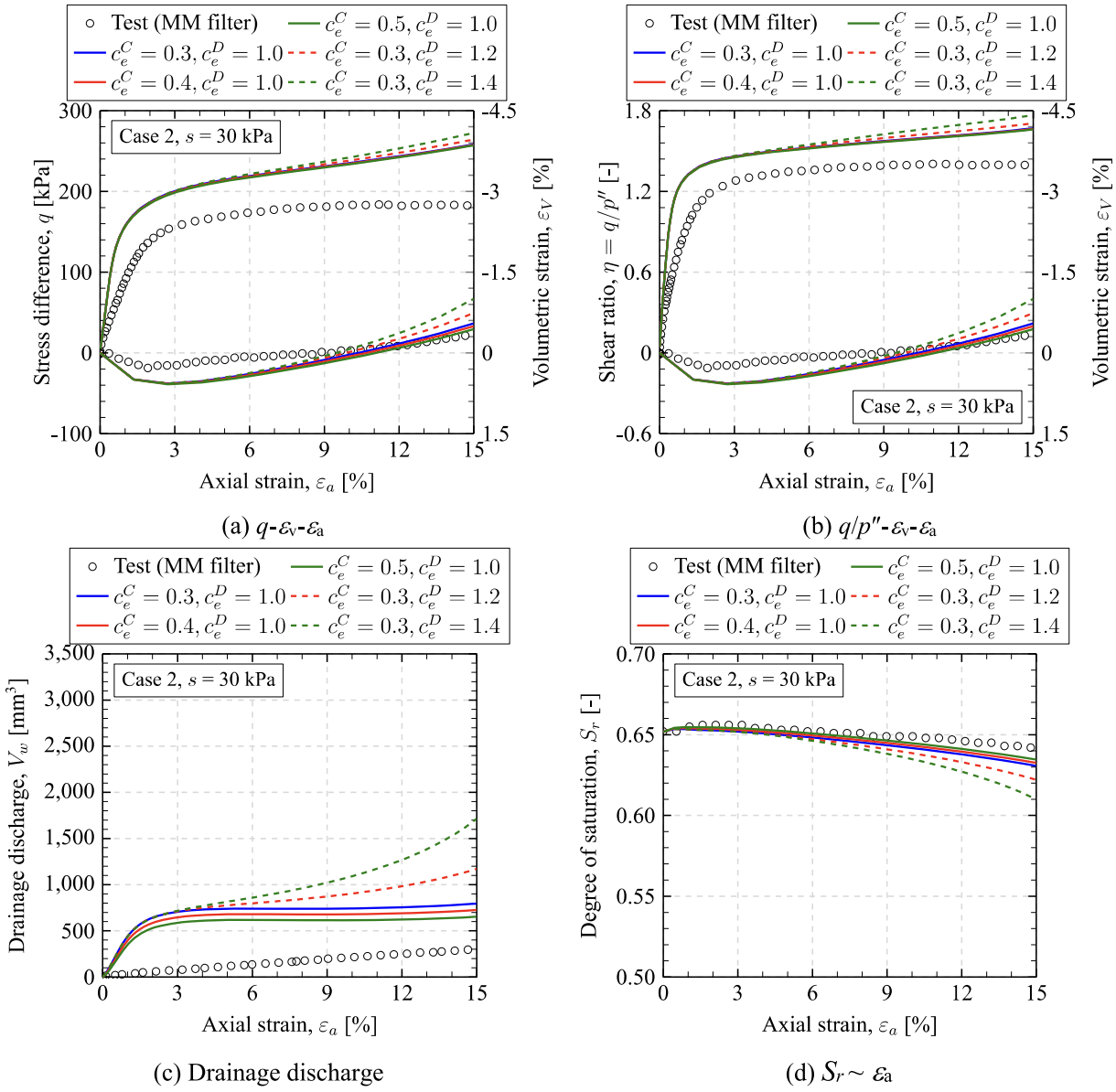
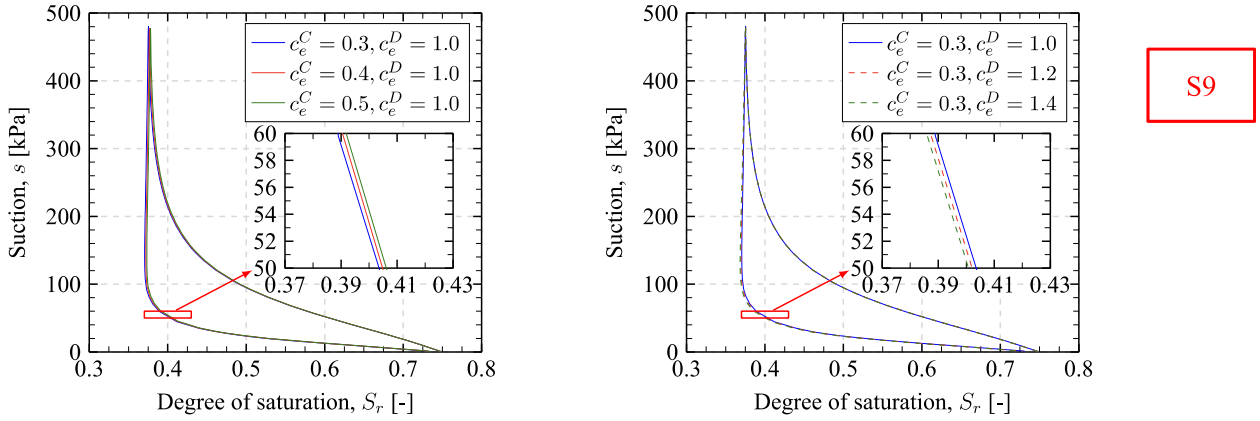


Fig. 10. Influence of finite deformation parameter c_e on calculated results (MM filters).

played to further investigate the influence of the ceramic disks and MM filters.

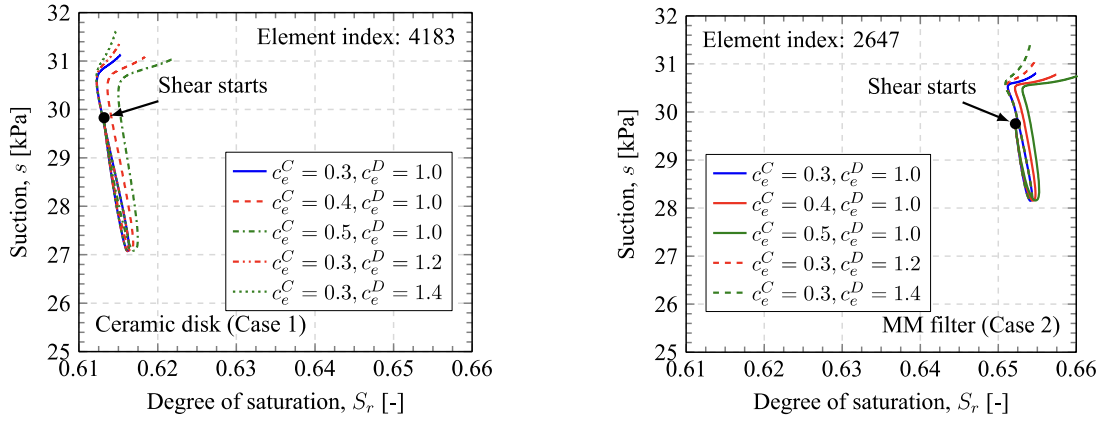
3.4. Determination of apparent triaxial test results

In order to compare the results of the element tests and corresponding FE-FD simulations, the following algorithm was adopted to derive the apparent triaxial test results. Considering the horizontal restrictions between each ceramic disk/MM filter element and the soil element, the elements along the middle layer of the specimen were subjected to vertical loads only to offer a more precise description of the mechanical behavior. In addition, the Bishop-type effective stress of the specimens differed by the layers. Thus, the axial load applied on each specimen was calculated by summing up the vertical loads of all the elements along the middle layer of the specimen. The stress difference (q) was firstly obtained by dividing the applied axial load by the sectional area of the middle layer of the specimen in the vertical direction, and then subtracting the confining stress. The stress ratio (η) was calculated by dividing the stress difference (q) by the mean skeleton stress (p") whose value was obtained by averaging the lateral skeleton stress of the middle layer elements along the ring boundary. The volumetric strain (ε_V) of the specimen was determined by dividing the total volume change of all the elements of Masado by the initial volume of each specimen. The amount of drainage discharge of all the soil elements was totaled in order to determine the drainage discharge (V_w) of each specimen.



S9

(a) Transition of WRC of Masado along with changes in c_e



(b) s-S_r relationship for same element when different parameters are used in FEM calculation

Fig. 11. Influence of finite deformation parameter c_e on WRC.

3.5. Influence of c_e on calculation results

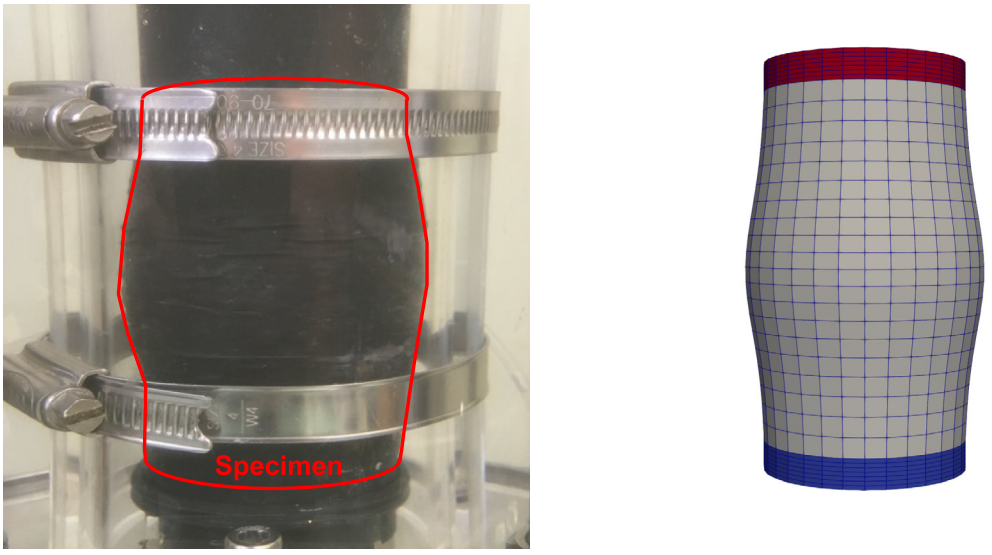
The constitutive model proposed by Xiong et al. (2021) indicated that the increment in saturation (dS_r) is composed of two parts: the adsorption and discharge of the soil (dS_r^s) and the change in void ratio (dS_r^e).

$$dS_r = dS_r^s + dS_r^e = k_s^{-1} ds + c_e de \tag{1}$$

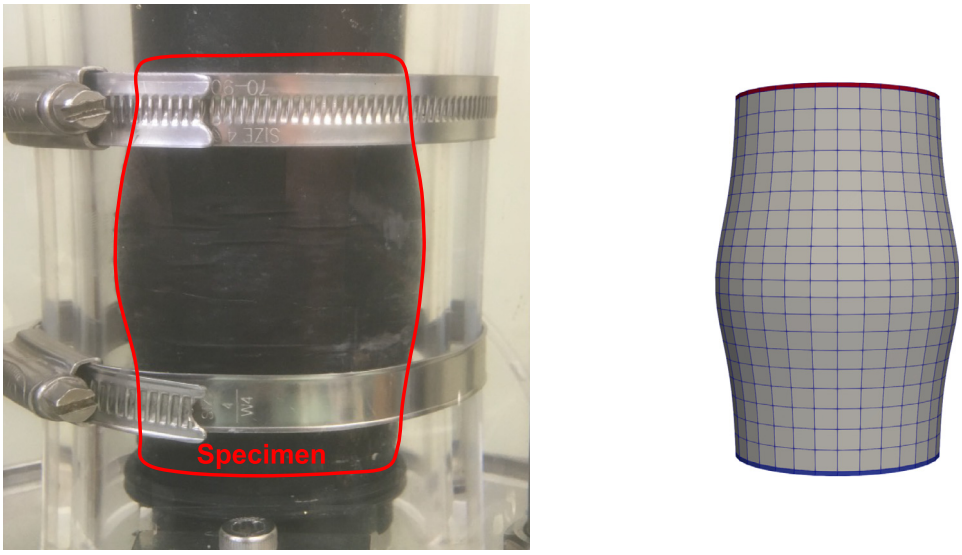
where k_s is the tangential stiffness of the suction-saturation curve, ds is the increment in suction, and de is the increment in void ratio. In addition, parameter c_e, called the deformation parameter (Xiong et al., 2021), controls the influence of deformation on the variation in the degree of saturation. The concept is that the variation in void ratio de linearly introduces a shifting value of dS_r on the intrinsic WRC along the S_r axis. Parameter c_e controls the length of this shift caused by unit de. Different values for c_e could be determined for compression (c_e^C) and dilatancy (c_e^D) conditions. Xiong et al. (2021) suggested that values for c_e should be determined by the curve fitting method based on the S_r-e-s relations obtained from water retention tests. The variation in c_e significantly influences the apparent behavior of the simulated Masado, which is discussed here in advance to achieve an in-depth and clear understanding of it.

Figs. 9 and 10 present the influence of finite deformation parameter c_e on the calculated results for the cases with the ceramic disks and MM filters, respectively. In the case of the ceramic disks, the strength of Masado, shown in Fig. 9(a), and the stress ratio, shown in Fig. 9(b), varied little regardless of the changes in c_e^C and c_e^D. Similar calculation results can be seen in the case of the MM filters shown in Fig. 10. As shown in Fig. 9(a) and Fig. 10(a), the increase in c_e^C led to a slight increase in ε_v during the shearing process, while the influence of c_e^D was prominent. A significant difference can be observed between Fig. 9(c) and Fig. 10(c), namely, that the volume of the discharge climbed to 1800 mm^3 in the case of $c_e^C = 0.3$, $c_e^D = 1.4$, which resulted in an obvious decline in S_r, as shown in Fig. 9(d) and Fig. 10(d).

In order to explain the differences in soil behavior due to the change in parameter c_e, the WRC of Masado was carefully investigated. Fig. 11(a) illustrates the transition in the WRC caused by the variation in parameter c_e. The S_r of Masado rose with the increase in c_e^C under constant suction, while parameter c_e^D resulted in an opposite variation. Similar results can be observed in the s-S_r relationship at



(a) Case 1: Ceramic disk



(b) Case 2: MM filter

Fig. 12. Comparison of deformed specimens between tests and calculation at end of shearing stage.

specified elements, as seen in Fig. 11(b). The selected element was in the center of the horizontal plane located in the middle of the specimen, as shown in Fig. 8. When the suction was kept constant, a higher value for the S_r of the specimen was observed with the increase in c_e^C, resulting in a larger quantity of water remaining in the specimen. Therefore, the amount of drainage discharge decreased, leading to an increase in the dilatancy of the specimen. The variation in S_r affected by c_e^D, shown in Fig. 11(b), was opposite to that affected by c_e^C. Therefore, an in-depth understanding of the parameter c_e involved in the WRC is important for comprehending the hydraulic properties of the soil material. With the above-mentioned observations and considerable complexity in obtaining the most suitable parameters, values of 0.3 and 1.0 were utilized in the comparison between the triaxial test results and the simulation with different ATTs for c_e^C and c_e^D, respectively. The determination of c_e^C and c_e^D do need further investigation in order to obtain the best results because of the significant difference in the hydraulic behavior of Masado caused by a small variation in parameter c_e. In addition, it should be noted that the determination of c_e^C and c_e^D would be quite difficult since the ceramic disks used in the unsaturated tests might lead to a large error in the measuring values of saturation shown in Fig. 5(d).

3.6. Elementary behavior results

Fig. 12 compares the deformed specimens of the triaxial tests and calculation results at the end of the shearing stage. The solid red line in the photos represents the contour of the specimen, which was covered by a rubber membrane. After the triaxial tests and numerical calculation, neither of the specimens displayed the characteristics of a

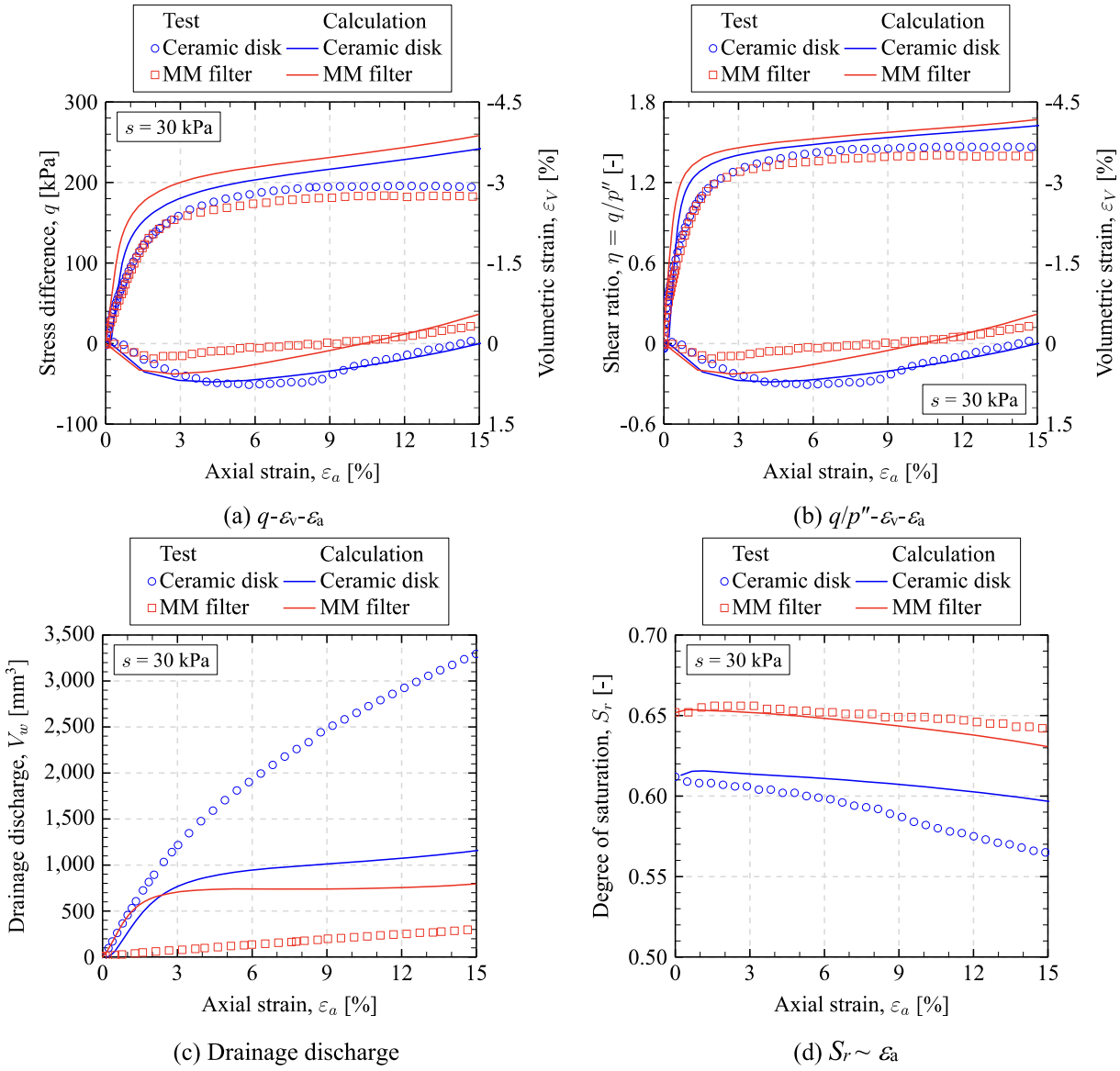


Fig. 13. Comparison of elemental behavior between tests and calculation in shearing stage.

shear band. The overall final shape of each specimen was reproduced by the simulation in both cases, namely, that of the ceramic disks and that of the MM filters.

Based on the above-mentioned calculation algorithm, a comparison of the elemental behavior between the triaxial tests and the numerical calculation is shown in Fig. 13. The results of the tests on Masado were basically reproduced by the simulation. The calculated stress–strain relationship in the case of the ceramic disks (Case 1) had the same tendency as that in the case of the MM filters (Case 2). The difference in stress increased continuously in both the tests and the calculation, while the initial stiffness of the stress–strain relationship and the difference in residual stress (q) obtained by the calculation were higher than those obtained by the triaxial tests, as shown in Fig. 13(a). The calculated stress ratios (η) in Cases 1 and 2 were also larger than those from the tests, as displayed in Fig. 13(b). Since the constitutive model itself was able to describe the element behavior of Masado to a large extent (Xiong et al., 2021), a considerable reason for this difference is the influence of the ceramic disks/MM filters on the specimen, or the non-uniform distribution of the so-called element behavior supposed to be in the tests. A detailed explanation of this reason will be provided later.

A satisfactory simulation of the volumetric strain, ε_V, can also be observed for Cases 1 and 2, as shown in Fig. 13(a). Calculated ε_V transformed from compression to dilatancy, which was in agreement with the test results. Moreover, the phase transitions calculated in Cases 1 and 2 corresponded with the test results, and the calculation reached an identical volumetric strain in the tests at $\varepsilon_a = 15$ %.

Furthermore, the drainage discharge given in Fig. 13(c) reproduced the difference in drainage between Case 1 and

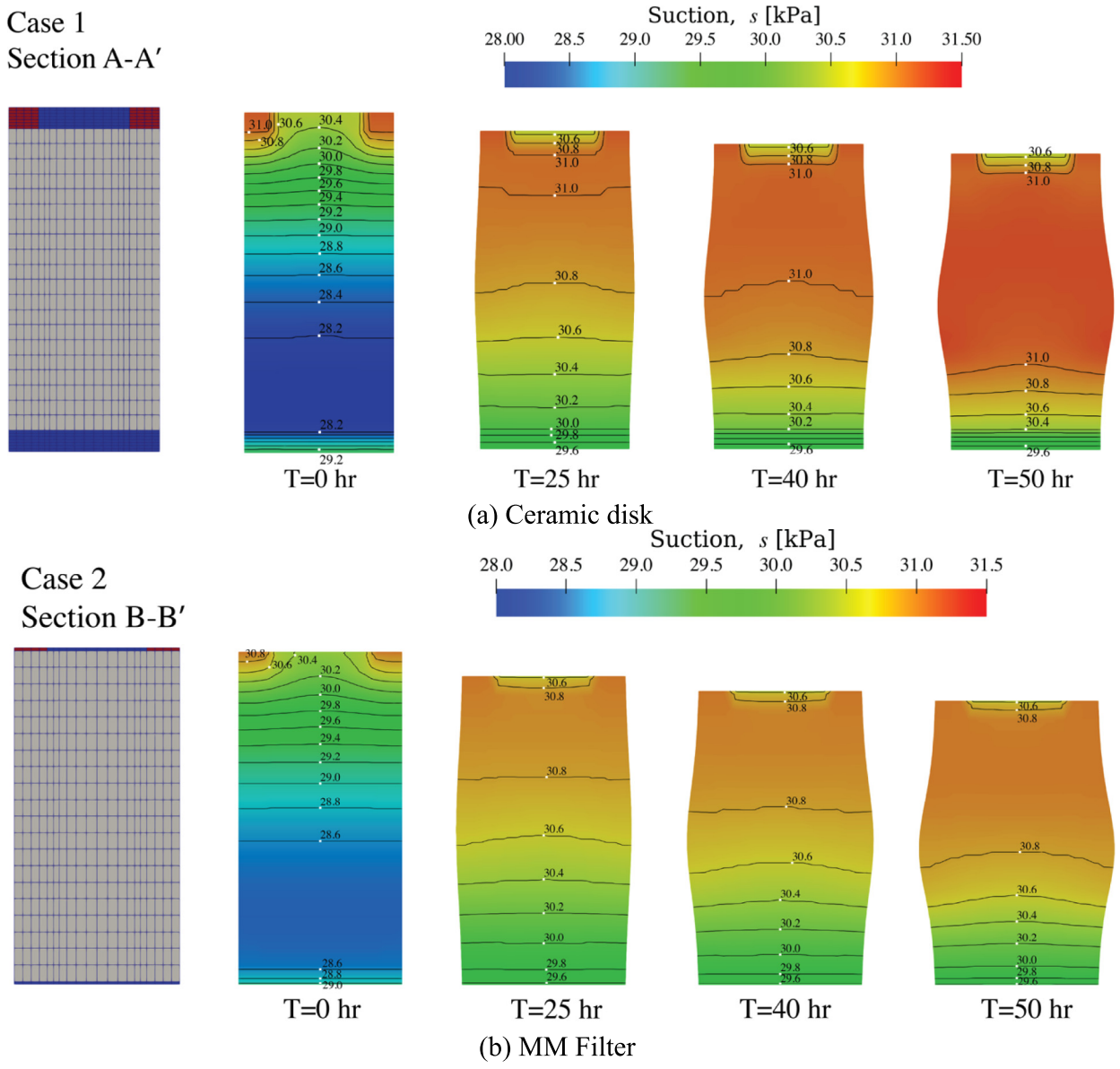
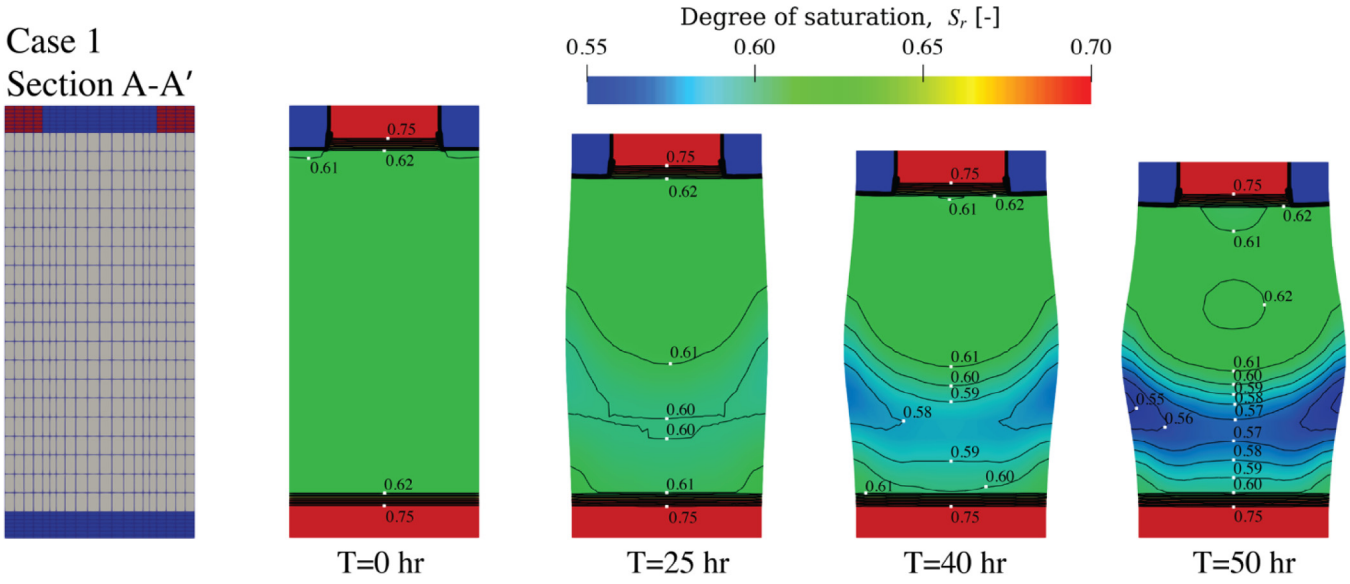


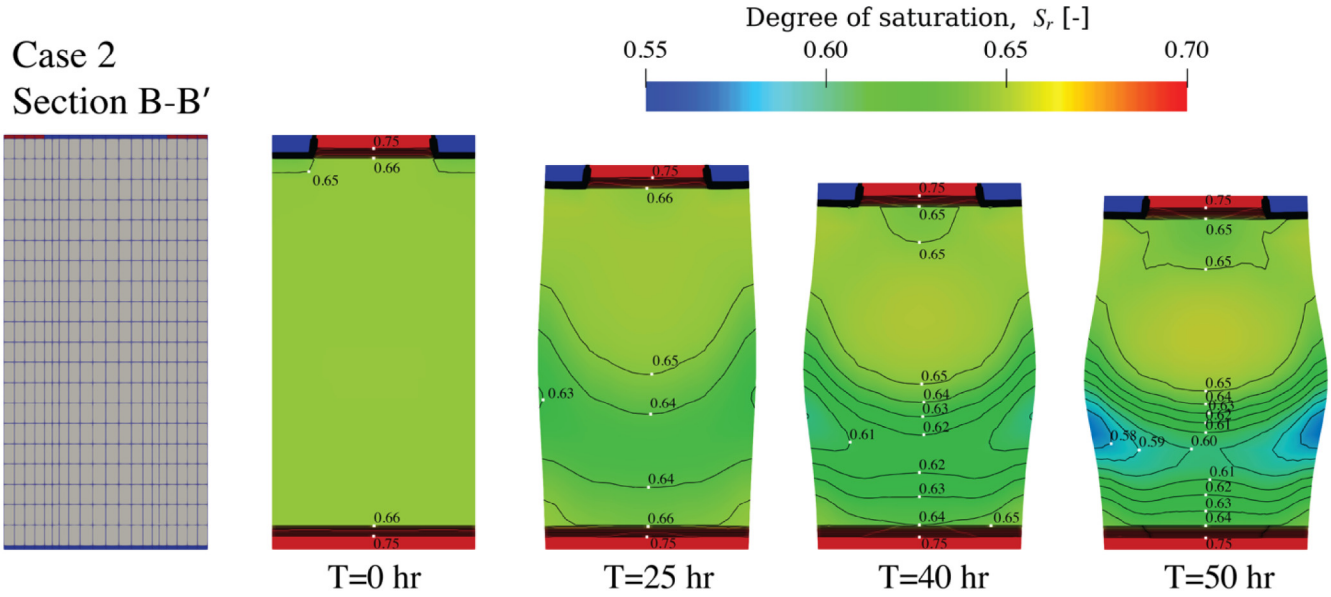
Fig. 14. Calculated distributions of suction at specified times in shearing stage.

Case 2, namely, that a larger quantity of water was discharged from the specimens in the case of the ceramic disks than in the case of the MM filters. The calculation verified the drainage difference observed during the triaxial tests and confirmed the influence of the materials for the ATT on the drainage behavior of the specimens. However, it can be observed in Fig. 13(c) that V_w in Case 1 in the triaxial tests was about 2200 mm^3 larger than that in the calculation, while V_w in Case 2 during the triaxial tests was 300 mm^3 smaller than that in the calculation. In other words, the difference in V_w between Case 1 and Case 2 in the calculation was not as large as that obtained from the tests. The following reasons may be responsible for this discrepancy. The thickness of the MM filters was 1 mm in the mesh (Fig. 8(b)), while the thickness of the MM filters used in the triaxial tests was only 125 μm in reality. In addition, the WRC took into consideration the influence of the finite deformation on the degree of saturation. The hydraulic and mechanical properties were so very highly coupled that an inevitable difficulty was encountered in the simulation due to the intertwining properties.

The influence of the soil deformation on S_r is shown in Fig. 13(d). After a slight increase in saturation at the beginning of the shearing stage, the calculated S_r decreased with the increase in ε_a. A similar variation in S_r was also observed in the triaxial tests in the case of the MM filters. However, the decrease in S_r at the end of shearing in the simulation ran contrary to that in the



(a) Ceramic disk (Test: before compression S_r of Masado =0.613; end of compression S_r =0.562)



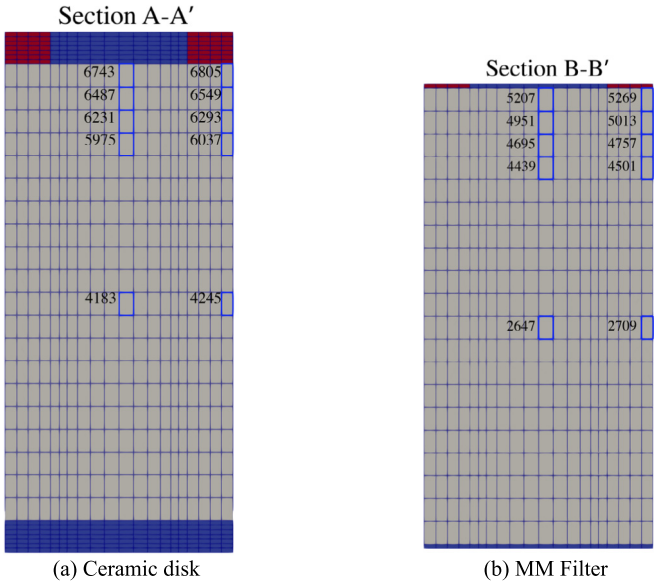
(b) MM filter (Test: before compression S_r of Masado =0.652; end of compression S_r =0.639)

Fig. 15. Calculated distributions of degrees of saturation at specified times in shearing stage.

tests. In the calculation, the dilatancy in Case 2, shown in Fig. 13(a), was larger than that in Case 1. The dilatancy of the specimens caused a reduction in the saturation of the specimens. Therefore, even though the calculated discharge in Case 2 was smaller than that in Case 1, the variation in S_r in Case 1 did not show any obvious difference compared to Case 2. The calculation was conducted under the constant suction condition (the water pressure and air pressure set on the boundaries were fixed); therefore, the variations in the degree of saturation and the drainage discharge were accompanied by changes in the void ratio. These calculated results indicate that the soil deformation significantly influences the hydraulic behavior of soil specimens, highlighting the importance of finding a suitable c_e parameter.

3.7. Distribution of suction and saturation

The calculated distributions of suction at specified times in the shearing stage are shown in Fig. 14. The calculated suctions in Cases 1 and 2 were roughly evenly distributed during the shearing process. The maximum difference in suction between the specimens was less than 1 kPa, indicating that the simulation was identical to an ideal triaxial test under constant suction conditions. At the end of the tests (50 h), the suction of many elements within the specimen,

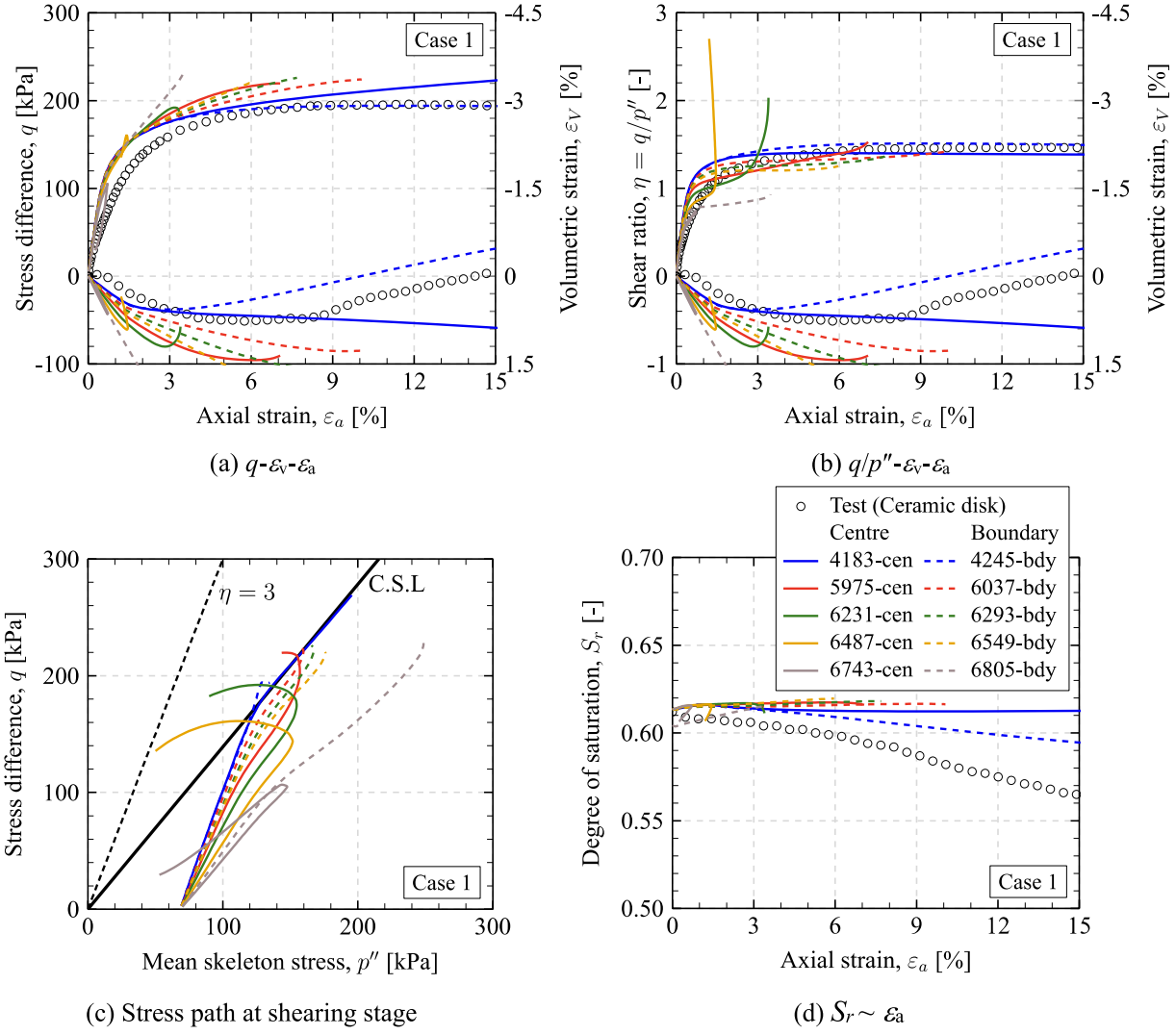


(a) Ceramic disk (b) MM Filter

Fig. 16. Specified elements to display stress and deformation properties.

in the case of the ceramic disks, was over 31.0 kPa, which was larger than the elements in the specimens in the case of the MM filters. This is thought to be the reason why a larger quantity of water was discharged from the specimens in the case of the ceramic disks than in the case of the MM filters (Fig. 13(c)). Furthermore, considering that the suction applied to the specimens was 30 kPa, the calculated suction in Case 2 was closer to the prescribed suction. The difference in suction distributions between the MM filter case and the ceramic disk case, although with an acknowledged small discrepancy, indicates that MM filters show the potential to perform better in controlling suction than ceramic disks.

Obvious non-homogeneous distributions of S_r can be seen at specified times in the shearing stage, as shown in Fig. 15. S_r decreased gradually as the shearing progressed. The soil elements near the ceramic disks or MM filters were deeply affected by the high value of S_r, leading to non-uniform distributions of S_r. In addition, it can be observed



(a) q-ε_v-ε_a (b) q/p''-ε_v-ε_a

(c) Stress path at shearing stage (d) $S_r \sim \varepsilon_a$

Fig. 17. Calculated responses of specified elements near ceramic disks.

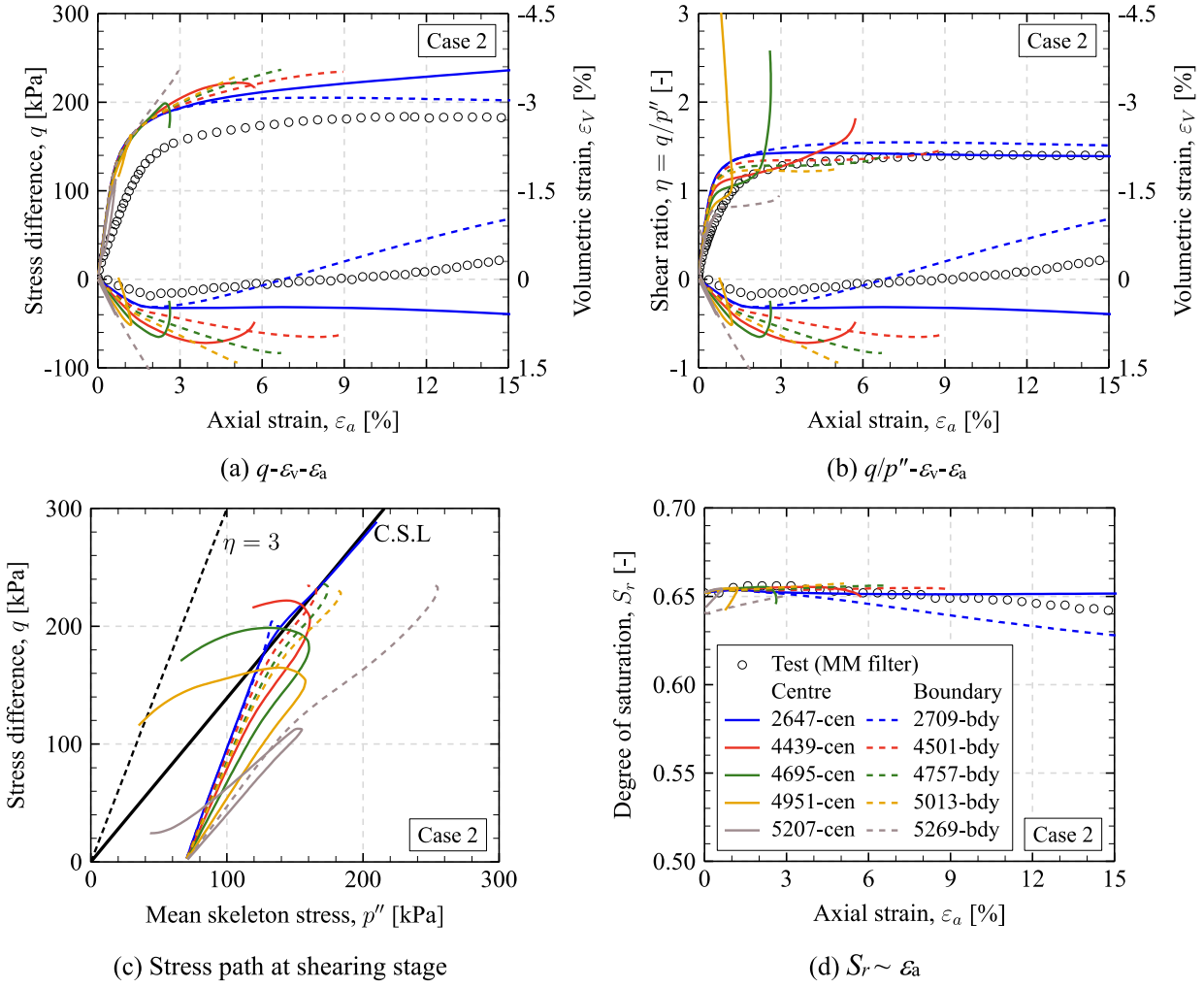


Fig. 18. Calculated results of specified elements near MM filters.

from Fig. 14 that the higher suction in the case of the ceramic disks could result in a lower value for S_r than that in the case of the MM filters. Moreover, the discrepancy in volumetric deformation between the two cases cannot be ignored. In another words, the triaxial tests using the ceramic disks/MM filters cannot be considered as strict element tests. Thus, the behavior of specific Masado elements during the shearing process needs to be analyzed.

3.8. Influence of ceramic disks and MM filters on Masado element behavior

As mentioned above, calculated stress difference q, given in Fig. 13(a), and the stress ratio, given in Fig. 13(b), can qualitatively, but not fully quantitatively, reproduce the triaxial test results. One possible reason is that the elements near the ceramic disks or MM filters are greatly restrained by the disks or filters. To verify this assumption, special attention was paid to the elements shown in Fig. 16 in order to investigate their mechanical behavior.

Referring to the selected elements shown in Fig. 16(a) for the ceramic disks, Fig. 17 illustrates their calculated responses during the shearing stage. The solid lines, denoted by 'cen', stand for the elements at the center of the horizontal layer, while the elements on the boundary of the specimen are denoted as 'bdy' and use dashed lines for their curves. It is obvious from Fig. 17 that the hydro-mechanical behavior of the elements is closely related to their locations in the specimen. It is appropriate to consider triaxial tests using ceramic disks/MM filters in the ATT as BVPs. Moreover, the elements at the center of the layer exhibit a much larger stress ratio than the elements along the boundary, as shown in Fig. 17(b). The skeleton stress path (q-p") in the shearing stage, shown in Fig. 17(c), demonstrates that the central elements near the ceramic disk (element indices: 5975, 6231, and 6487) gradually approach the dashed line ($\eta = 3$). Meanwhile, the stress paths for the elements along the boundary near the ceramic disk (element indices: 6037, 6293, and 6549) finally approach the CSL. The stress ratio will not equal 3 unless the effective lateral stress (σ_3'') equals 0. Except for the element next to the ceramic disk (element index: 6743), the closer the elements along the vertical center axis of the specimen are to the ceramic disk, the smaller the lat-

eral skeleton stress will be. However, the stress paths of the elements at the middle of the specimen (element indices: 4183 and 4245) were basically in agreement with the unsaturated triaxial test results compared to the other selected elements, indicating the negligible influence of ceramic disks on these elements.

The characteristics of the hydro-mechanical behavior of elements in the case of MM filters are similar to those in the case of ceramic disks, as shown in Fig. 18. A negative σ_3'', caused by a continuous decrease, would result in a η $(=3(\sigma_1'' - \sigma_3'')/(\sigma_1'' + 2\sigma_3''))$ over 3. Thus, element 4951 may even be under the tensile state in the radial direction!

In summary, the calculated results of the specified elements located near a ceramic disk or MM filter can explain the discrepancy in the stress differences given in Fig. 13. The ceramic disk or MM filter is responsible for the uneven distribution of the mechanical responses of the soil elements, resulting in the inconsistency between the calculated elementary results and the triaxial test results. This phenomenon refers not only to the displacement restriction between the soil specimens and the ATT materials, but also to the drainage discrepancy induced by the existence of ATTs.

4. Conclusions

In this study, the hydraulic-mechanical behavior of Masado was investigated through unsaturated triaxial tests. The results were simulated by a soil–water-air coupling analysis with the FE-FD method to investigate the influence of ceramic disks/MM filters and the non-uniform deformation of the specimens. Firstly, the ATTs with the ceramic disks/MM filters were utilized to control the matric suction in triaxial tests on unsaturated Masado. Then, based on the unsaturated/saturated constitutive model coupled with a deformation-dependent WRC model, proposed by Xiong et al. (2021), numerical simulations were conducted with the soil–water-air coupling FE-FD method to simulate the triaxial tests, by regarding the element tests as BVPs. The material parameters of the constitutive model and the WRC model were determined based on the method by Xiong et al. (2021). Finally, the calculated results were compared with the triaxial test results to verify the performance of the constitutive model in describing the hydro-mechanical properties of Masado. The main conclusions of this study are presented as follows:

(1) The triaxial test results using ceramic disks or MM filters showed little difference in terms of the stress–strain relation. In both cases, the volumetric strain transformed from compression to dilatancy. However, a significant discrepancy between the two methods occurred mainly in the S_r variation. The amount of drainage discharge during shearing in the case of the ceramic disks was much larger than that in the case of the MM filters.

(2) The stress–strain relation of Masado obtained from FE-FD calculation was able to appropriately reproduce the test results. By comparing the strength and the stress path of specified elements, it was found that both ceramic disks and MM filters can significantly influence the mechanical behavior of the Masado elements located near them. The uneven distributions of the deformation and strength of the soil within the specimens, in the triaxial tests using both ceramic disks and MM filters, proved that the so-called element test is not an elementary problem, but a BVP.

(3) The calculations conducted with the FE-FD method successfully and qualitatively reproduced the difference in drainage discharge from the specimens using the ceramic disks/MM filters, but not fully quantitatively. The calculated discrepancy was much less than that of the tests. The reason for this discrepancy was caused by the sensitive change in the deformation parameter, c_e, on the WRC, because the simulation of the hydraulic-mechanical behavior of Masado was deeply coupled, leading to the difficulty of a precise numerical calculation. Therefore, the determination of c_e in compressive and dilatant states should be carefully investigated since it significantly influences the unsaturated behavior of soils.

(4) The distribution of suction in the specimens was observed to be generally uniform throughout the shearing tests regardless of the ATTs, while S_r was found to be unevenly distributed in both cases. Lower saturation was mainly concentrated in the middle of the specimen because the high value of S_r of the ceramic disks/MM filters significantly raised the saturation of the elements near them. However, higher suction was observed overall in the case of the ceramic disks compared to that of the MM filters, resulting in a larger drainage discharge during the shearing stage. Considering that the calculated suction with the MM filters was closer to the prescribed suction applied to the specimens, MM filters are concluded to be superior to ceramic disks in controlling the suction during tests.

(5) In the numerical simulations, it was shown that, although the tests conducted in this research are BVPs, not elementary problems, the average behavior of the specimens calculated by the numerical simulation was basically the same as in the triaxial test results which were assumed to reflect the element behavior, indicating that triaxial tests with ATTs using ceramic disks/MM filters can basically be used to verify the hydraulic-mechanical behavior of geomaterials.

Acknowledgements

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Appendix A. Brief introduction of constitutive model

Zhang and Ikariya (2011) proposed a saturated/unsaturated constitutive model to model the hydro-mechanical soil behavior. The constitutive model employed the skeleton stress and degree of saturation as the state variables. The skeleton stress is defined as

$$\sigma''_{ij} = \sigma^t_{ij} - u_a\delta_{ij} + S_r(u_a - u_w)\delta_{ij} = \sigma^n_{ij} + S_r s\delta_{ij} \tag{A-1}$$

In the constitutive model, the concepts of subloading (Hashiguchi and Ueno, 1977) and superloading (Asaoka et al., 2002) were adopted to describe the overconsolidation and structure of the soil. The similarity ratio of the superloading yield surface to normal yield surface R^* and the similarity ratio of the superloading yield surface to subloading yield surface R are given as

$$R^* = \frac{p^*}{p'} = \frac{q^*}{q'}, \quad 0 < R^* \le 1 \quad \frac{q^*}{p^*} = \frac{q'}{p'} \tag{A-2}$$

$$R = \frac{p}{p'} = \frac{q}{q'}, \quad 0 < R \le 1, \quad R = \frac{1}{OCR} \quad \frac{q}{p} = \frac{q^*}{p^*} = \frac{q'}{p'} \tag{A-3}$$

where, $(p,\ q)$, $(p^*,\ q^*)$, and $(p',\ q')$ represent the present stress state, the corresponding normally consolidated stress state, and the structured stress state at the p-q plane, respectively. Then the yield function is given as follows:

$$f = \frac{p}{p_0} + ln\frac{\mathrm{M}^2 + \eta^2}{\mathrm{M}^2} + \frac{\rho_e}{1+e_0}\frac{1}{C_p} - \frac{\rho^*}{1+e_0}\frac{1}{C_p} - \frac{\rho_s}{1+e_0}\frac{1}{C_p} - \varepsilon^p_v\frac{1}{C_p} = 0 \tag{A-4}$$

where

$$C_p = \frac{\lambda-\kappa}{1+e_0} \quad \rho^* = (\lambda-\kappa)ln\frac{1}{R^*}$$
$$\rho_s = N(S_r) - N \quad \rho_e = (\lambda-\kappa)ln\frac{1}{R} \tag{A-5}$$

where λ, κ, and M are the compression index, swelling index, and stress ratio at the critical state, respectively. p_0 is the reference mean skeleton stress. ε^p_v is the plastic volumetric strain. ρ^* represents the difference in void ratio between the normally consolidated state and the structured state under the same mean skeleton stress. ρ_s represents the difference in void ratio between the normally consolidated line in the saturated state and the unconsolidated line under the same skeleton stress. $N(S_r)$ and N are the void ratio at a specific value of S_r and at the fully saturated state under the reference mean skeleton stress, respectively. ρ_e represents the difference in void ratio between the normally consolidated state and the overconsolidated state under the same mean skeleton stress.

In this constitutive model, the following associated flow rule is adopted:

$$d\varepsilon^p_{ij} = \Lambda\frac{\partial f}{\partial\sigma_{ij}} \tag{A-6}$$

where Λ is the plastic scalar factor.

From the consistency equation, $\mathrm{d}f = 0$, the consistency equation for the yield surface can be given as

$$df = \frac{\partial f_\sigma}{\partial\sigma_{ij}}d\sigma_{ij} + d\left(\frac{\rho_e}{1+e_0}\right)\frac{1}{C_p} - d\left(\frac{\rho^*}{1+e_0}\right)\frac{1}{C_p} - d\left(\frac{\rho_s}{1+e_0}\right)\frac{1}{C_p} - d\varepsilon^p_v\frac{1}{C_p} = 0 \tag{A-7}$$

The evolution equations for the development of state variables ρ^* of the structure, ρ_s of overconsolidation, and ρ_e of saturation are given as follows:

$$d\left(\frac{\rho^*}{1+e_0}\right) = e^{b\rho_s}\left(1 - e^{m^*\rho^*}\right)\Lambda\frac{1}{\sigma_{mm}} \tag{A-8}$$

$$d\left(\frac{\rho_e}{1+e_0}\right) = -\Lambda\frac{\rho^\beta}{\sigma_{mm}}, \rho = a\rho_e + b\rho_s \tag{A-9}$$

$$\begin{cases} N(S_r) = N + \frac{N_r - N}{S^s_r - S^r_r}\left(S^s_r - S_r\right); N_r = N\left(S^r_r\right) \\ \rho_s = N(S_r) - N = Q\left(S^s_r - S_r\right); Q = \frac{N_r - N}{S^s_r - S^r_r} \\ d\rho_s = -QdS_r \end{cases} \tag{A-10}$$

where m^* is the parameter controlling the rate of collapse of the structure during shearing. S_r^{s} and S_r^{r} are the degrees of saturation under residual and saturated conditions, respectively. Parameters a, b, and β control the evolution of state variables ρ^* and ρ_e, and N_{r} is the void ratio at the normal consolidation line under reference mean skeleton stress p_0 when the degree of saturation is in the residual state.

The positive variable Λ can be rewritten as

$$\Lambda = \frac{\frac{\partial f}{\partial\sigma_{ij}}d\sigma_{ij} + \frac{1}{C_p}\frac{Q}{1+e_0}dS_r}{\frac{1}{C_p}\frac{\rho^\beta}{\sigma_{mm}} + \frac{1}{C_p}e^{b\rho_s}\left(1 - e^{m^*\rho^*}\right)\frac{1}{\sigma_{mm}} + \frac{1}{C_p}\frac{\partial f}{\partial\sigma_{mm}}} = \frac{\frac{\partial f}{\partial\sigma_{ij}}E_{ijkl}d\varepsilon_{kl} + \frac{1}{C_p}\frac{Q}{1+e_0}dS_r}{\frac{h_p}{C_p}} \tag{A-11}$$

where

$$h_p = \frac{\rho^\beta + e^{b\rho_s}(1 - e^{m^*\rho^*})}{\sigma_{mm}} + \frac{\partial f}{\partial\sigma_{mm}} \tag{A-12}$$

The loading critical is given below.

$$\begin{cases} \Lambda > 0 & Loading \\ \Lambda = 0 & Neutral loading \\ \Lambda < 0 & Unloading \end{cases} \tag{A-13}$$

Appendix B. Brief description of water retention curve model

Xiong et al. (2021) proposed a deformation dependency water retention model to describe the water retention properties of soil and coupled it into the constitutive model proposed by Zhang and Ikariya (2011). The incremental of the degree of saturation dS_r is obtained by

$$dS_r = dS_r^s + dS_r^e = k_s^{-1} ds + c_e de \tag{B-1}$$

where $dS_r^{\,s}$ is attributed to removing the water from the soil and $dS_r^{\,e}$ is attributed to the change in void ratio, k_s is the tangential stiffness of the suction-saturation curve, and c_e is a parameter corresponding to the change in void ratio. ds and de represent the increments in suction and void ratio, respectively.

The skeleton curves of the water retention model can be given as

(a) Main drying curve

$$S_r = S_r^s - \frac{2}{\pi}\left(S_r^s - S_r^r\right) arctan\left(\frac{e^{c_1 s} - 1}{e^{c_1 S_d}}\right) - c_e(e - e_0) \tag{B-2}$$

(b) Main wetting curve

$$S_r = S_r^s - \frac{2}{\pi}\left(S_r^s - S_r^r\right) arctan\left(\frac{e^{c_2 s} - 1}{e^{c_2 S_w}}\right) - c_e(e - e_0) \tag{B-3}$$

where c_1 and c_2 are the shape function parameters controlling the intrinsic shape of the curves. $S_r^{\,s}$ and $S_r^{\,r}$ are the degrees of saturation under saturated and residual conditions, respectively. S_d and S_w are parameters corresponding to the drying air entry value and the wetting air entry value, respectively.

The incremental relation between suction and saturation of the scanning curve is expressed as follows:

$$k_s = \begin{cases} k_w + k_{dmax} r^{I_d} & ds \geq 0 \\ k_w + k_{dmax} r^{I_w} & ds < 0 \end{cases} \tag{B-4}$$

where k_w is the gradient of the corresponding main wetting curve under the same suction condition and $k_{d\max}$ is the gradient of the main drying curve under the residual condition. I_d and I_w are scaling factors that control the curvature of the scanning curve in the drying and wetting processes, respectively. The inner variable r is defined as

$$r = s / s_{max} \tag{B-5}$$

where $s_{\max}$ is the suction under the residual condition.

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